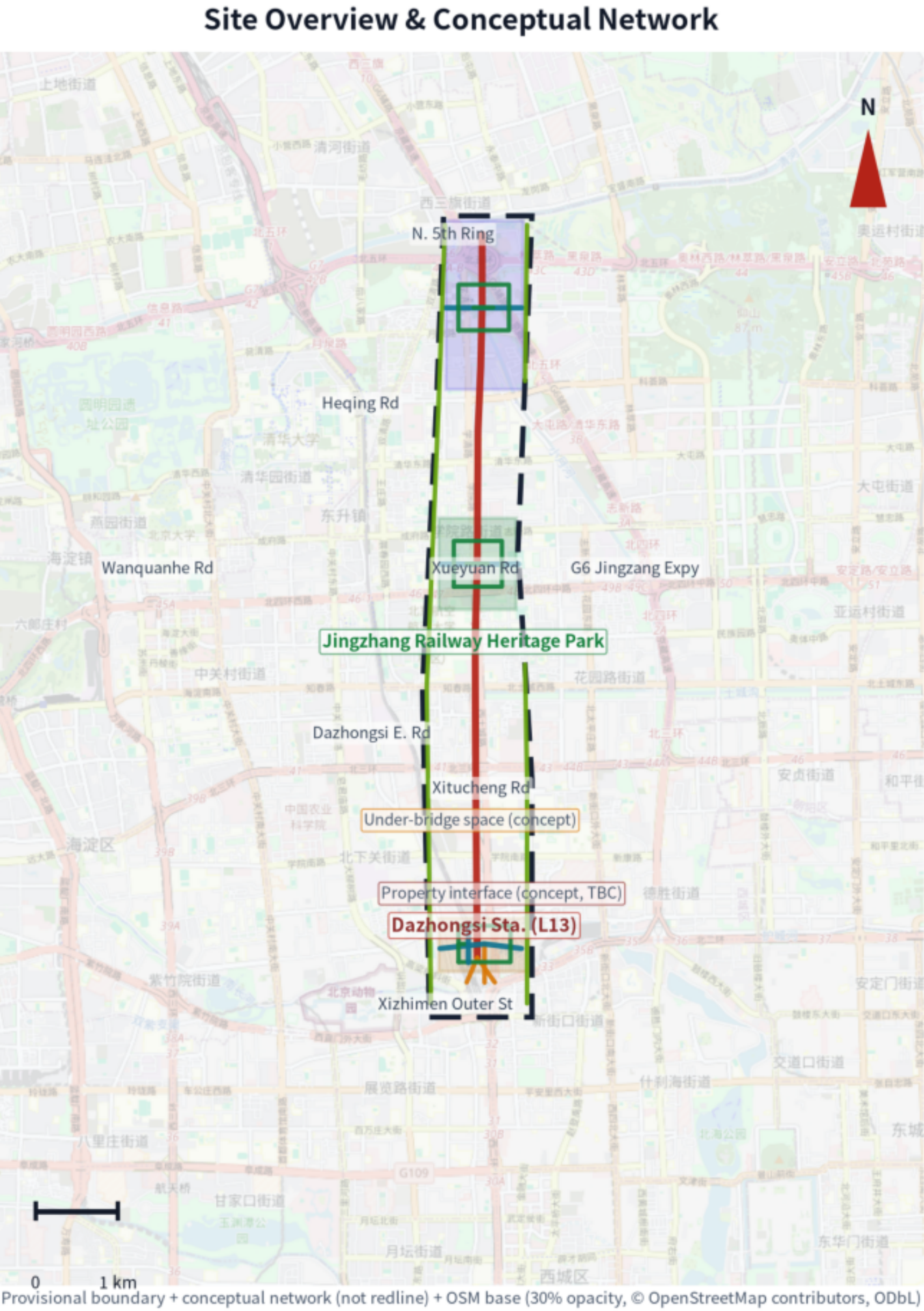


Bio-Physarum Road-Network Renewal for the Dazhongsi AI Cluster • Master Plan



Legend

- Physarum primary vein
- Secondary branch
- Local loop
- Jingzhang green corridor
- Station feeder
- Zhongzhiyuan Area
- Beijing AI Origin Community
- Dazhongsi AI Industry Cluster

Boundary/network/districts are algorithmic conceptual & provisional extents (not surveyed, not approval geometry). Road/station/park names are based on OSM public data and the brief's textual extents (illustrative). Under-bridge space and property interface are conceptual annotations, pending official verification.

19.20

Optimal eff.

2.802

Run7 frozen

8813 m

Skeleton length

17/18

Node coverage

80.34

Plan03 UDS

69.1%

Permeable

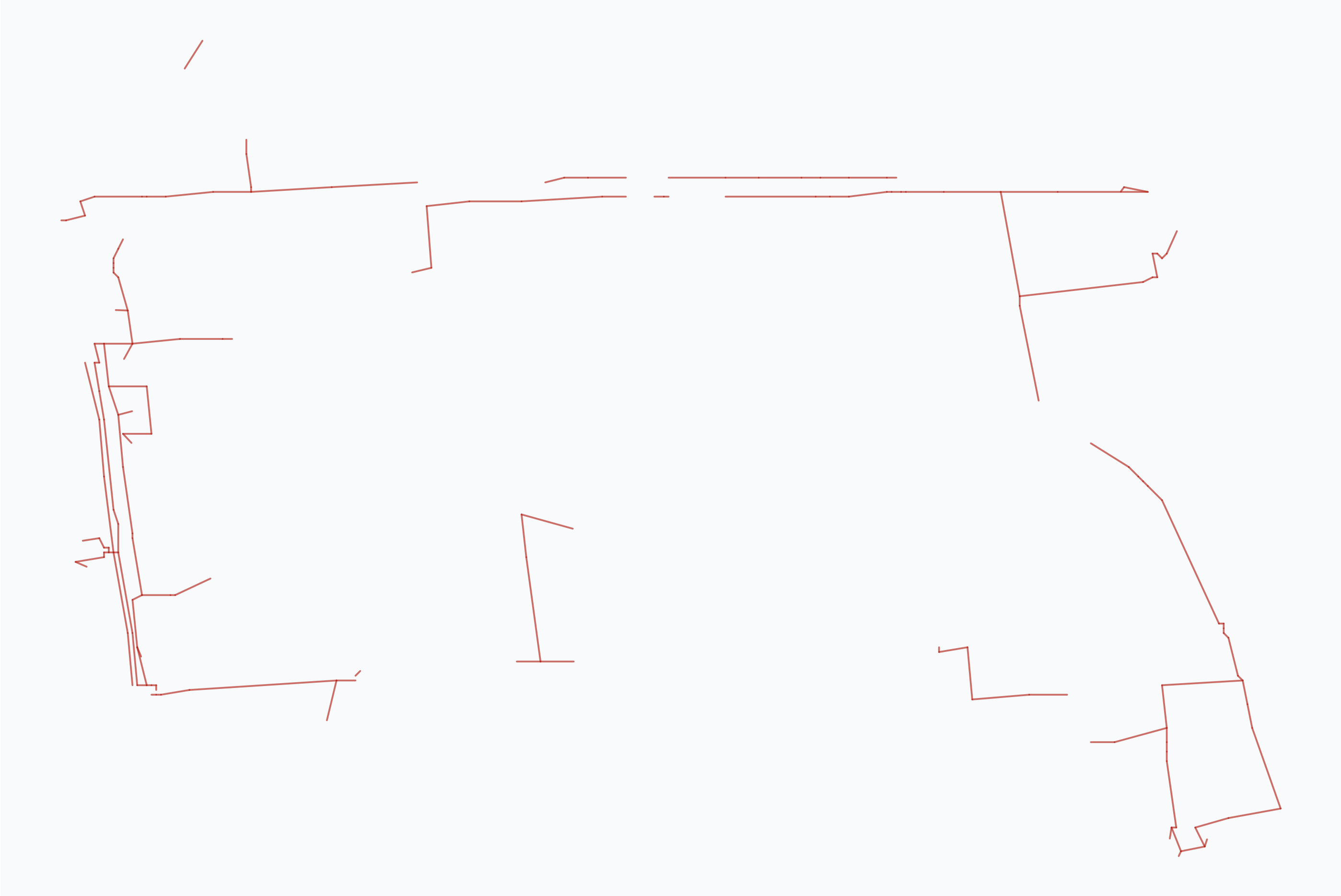
25.2%

Green pen.

Provisional warning: network / key areas / boundary are algorithm-generated concepts & provisional ranges (not surveyed, not approval redline); station coords = OSM public data; heritage-protection geometry unavailable — no redline drawn.

Method: Bio-Physarum + NSGA-II

Real Physarum 167-edge skeleton (method validation)



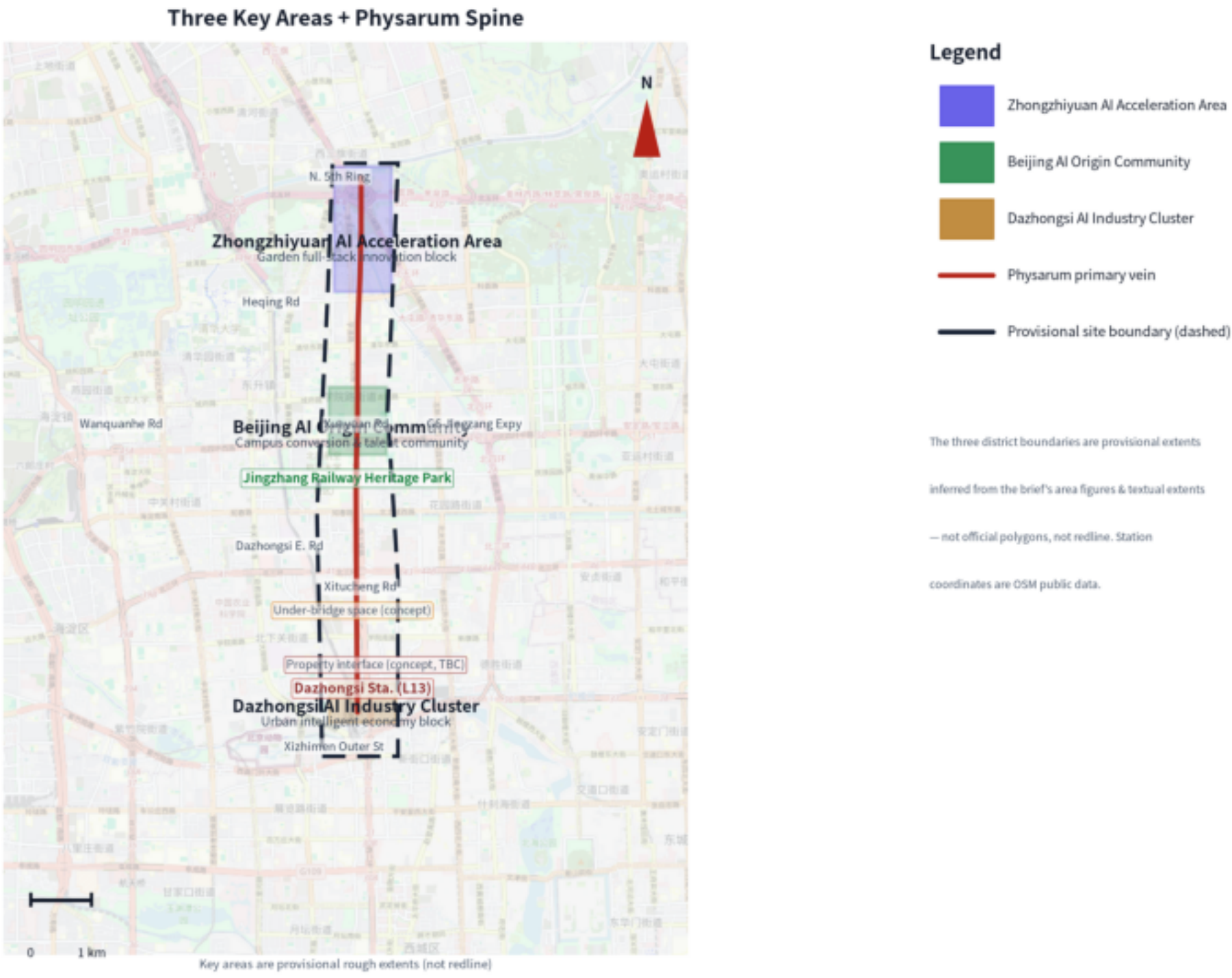
Metrics recalculation (same level • off-site validation)

OFF-SITE • not formal site geometry

0	19.20	167
Heritage hard-crossings	Optimal efficiency	Network edges
g1 hard constraint • off-site	Run7 frozen 2.802 • baseline 1.143 • off-site	real Physarum skeleton • off-site

Method validation (real Physarum run): 167-edge network / efficiency 19.20 / heritage objective f3≡f2 (degenerates to cost; no independent site heritage conclusion)

Key Areas & Global Cases

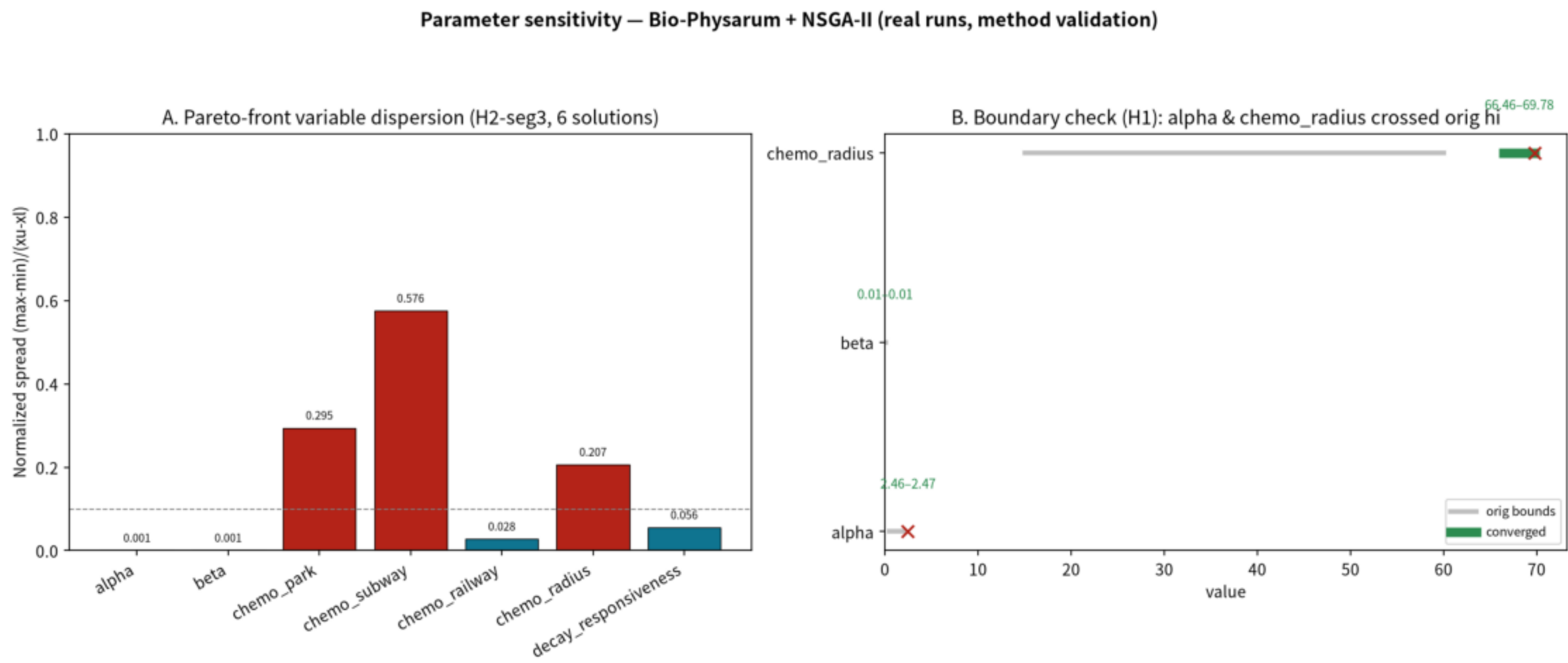


Global Benchmark Cases

案例	地点	核心做法（公开报道）	借鉴/警示
Sidewalk Labs Quayside	Toronto	Sensing, data trust (2017-2020)	Governance / caution
Masdar City	Abu Dhabi	Low-emission, passive cooling, PRT	Low-carbon + feeder
Songdo IBD	Incheon	Reclaimed town, U-City sensing	New infra reserve
Amsterdam	Netherlands	PPP, living-lab	Incremental + governance
Xiong'an	Hebei	CIM twin, blue-green, tunnels	Twin + blue-green

Test Scenarios & Implementation Matrix

Rigor & Implementability (Data Anchors)



Cost Breakdown (suggestive)			
等级	长度(m)	单价(元/m)	造价(万元)
Arterial	740.2	8000	592.2
Secondary	1744.1	5000	872.0
Branch	3679.9	3000	1104.0
Micro	2648.8	1500	397.3
Total	8813.0	—	2965.5

Avg 3365 CNY/m; municipal road only, excl. utility/rail/demo (suggestive)

Policy Alignment (suggestive)		
项目	政策/标准依据（真实）	待确认
JZ-01	GB 50763-2012 / barrier-free reg	ownership
JZ-02	Sponge guide / GB 50014	blue line
JZ-03	Beijing renewal reg (suggestive)	clauses
JZ-04	GB/T 51328 / GB 50763	station
JZ-05	New-infra policy (to confirm)	regulator
JZ-06	京政发〔1984〕128号	map/redline

